Summary:

Our project stores performance statistics for League of Legends champions, including win rate, pick rate and ban rate. It helps players make informed in-game decisions by offering insights into champion performance.

Changes of the ER diagram:

We used more specific attribute names across different entities to avoid confusion.

Schema:

Amateur_player(UID: char, Rank: char, Preferred_role: string)

Esports_player(UID: char, Rank: char, Team: string, Position: string)

Plays(**UID**: char, **ChampionName**: string)

Champions(ChampionName: string, Tier: char, Archetype: string, **Winrate**: double, **Pickrate**: double), winrate and pickrate can't be null

Stats(Winrate: double, Pickrate: double, Banrate: double)

Role(RoleName: string, Map_area: string)

Champions-In-Role(**RoleName**: string, **ChampionName**: string)

Matchup-during(Champion1: string, Champion2: string, Matchup Winrate: double, Matches Played: integer, **PatchID**: char)

Champions-Toughest-Matchup(**ChampionName**: string, **Champion1**: string, **Champion2**: string)

Build(SkillOrder: char, **ChampionName**: string, **date**: char)
Item(ItemName: string, Effect: string)

Build-Uses-Item(**SkillOrder**: char, **ChampionName**: string, **ItemName**: string)

Patch(PatchID: char, date: char)

FD:

Amateur_player:
UID -> Rank, UID -> Preferred_role

Esports_player:
UID -> Rank, UID -> Team, Position, Team -> Position

Role:
RoleName -> Map_area

Champions:
ChampionName -> Archetype, Tier
ChampionName, Tier -> Winrate, Pickrate

Stats:
Winrate, Pickrate -> Banrate

Matchup-during:
Champion1, Champion2 -> Matchup Winrate, Matches Played

Patch:
PatchID -> date

Item:
ItemName -> Effect

Champion1Name, Champion2Name, PatchID -> MatchupWinrate, MatchesPlayed

Normalization (3NF)

SQL DDL statements

```
CREATE TABLE User (
   UID   INTEGER PRIMARY KEY,
   Rank  VARCHAR(30)
);

CREATE TABLE Amateur_player (
   UID            INTEGER PRIMARY KEY,
   Preferred_role  VARCHAR(50),
   FOREIGN KEY (UID) REFERENCES User(UID) ON DELETE CASCADE
);

CREATE TABLE Esports_player (
   UID       INTEGER PRIMARY KEY,
   Team      VARCHAR(50),
   Position  VARCHAR(50),
   FOREIGN KEY (UID) REFERENCES User(UID) ON DELETE CASCADE
);

CREATE TABLE Champions (
   Name       VARCHAR(50) PRIMARY KEY,
   Archetype  VARCHAR(50),
   Tier       VARCHAR(10)
);

CREATE TABLE Stats (
   ChampionName  VARCHAR(50) PRIMARY KEY,
   Winrate       DECIMAL(5,2),
   Pickrate      DECIMAL(5,2),
   Banrate       DECIMAL(5,2),
   FOREIGN KEY (ChampionName) REFERENCES Champions(Name) ON DELETE CASCADE
);

CREATE TABLE Role (
   Name      VARCHAR(50) PRIMARY KEY,
   Map_area  VARCHAR(50)
```

```
);

CREATE TABLE ChampionInRole (
   RoleName      VARCHAR(50) NOT NULL,
   ChampionName  VARCHAR(50) NOT NULL,
   PRIMARY KEY (RoleName, ChampionName),
   FOREIGN KEY (RoleName) REFERENCES Role(Name) ON DELETE CASCADE,
   FOREIGN KEY (ChampionName) REFERENCES Champions(Name) ON DELETE CASCADE
);

CREATE TABLE Patch (
   PatchID  VARCHAR(10) PRIMARY KEY,
   date     DATE
);

CREATE TABLE Item (
   Name    VARCHAR(50) PRIMARY KEY,
   Effect  VARCHAR(255)
);

CREATE TABLE Build (
   ChampionName  VARCHAR(50) NOT NULL,
   SkillOrder    VARCHAR(20) NOT NULL,
   PRIMARY KEY (ChampionName, SkillOrder),
   FOREIGN KEY (ChampionName) REFERENCES Champions(Name) ON DELETE CASCADE
);

CREATE TABLE UserPlaysBuild (
   UID          INTEGER NOT NULL,
   ChampionName  VARCHAR(50) NOT NULL,
   SkillOrder    VARCHAR(20) NOT NULL,
   PRIMARY KEY (UID, ChampionName, SkillOrder),
   FOREIGN KEY (UID) REFERENCES User(UID) ON DELETE CASCADE,
   FOREIGN KEY (ChampionName, SkillOrder) REFERENCES Build(ChampionName, SkillOrder) ON
DELETE CASCADE
);

CREATE TABLE BuildUsesItem (
   ChampionName  VARCHAR(50) NOT NULL,
   SkillOrder    VARCHAR(20) NOT NULL,
   ItemName      VARCHAR(50) NOT NULL,
   PRIMARY KEY (ChampionName, SkillOrder, ItemName),
   FOREIGN KEY (ChampionName, SkillOrder) REFERENCES Build(ChampionName, SkillOrder) ON
DELETE CASCADE,
```

```
    FOREIGN KEY (ItemName) REFERENCES Item(Name) ON DELETE CASCADE
);

CREATE TABLE BuildInPatch (
    ChampionName  VARCHAR(50) NOT NULL,
    SkillOrder    VARCHAR(20) NOT NULL,
    PatchID       VARCHAR(10) NOT NULL,
    PRIMARY KEY (ChampionName, SkillOrder, PatchID),
    FOREIGN KEY (ChampionName, SkillOrder) REFERENCES Build(ChampionName, SkillOrder) ON
DELETE CASCADE,
    FOREIGN KEY (PatchID) REFERENCES Patch(PatchID) ON DELETE CASCADE
);

CREATE TABLE Matchup (
    Champion1Name   VARCHAR(50) NOT NULL,
    Champion2Name   VARCHAR(50) NOT NULL,
    PatchID         VARCHAR(10) NOT NULL,
    MatchupWinrate  DECIMAL(5,2),
    MatchesPlayed   INTEGER,
    PRIMARY KEY (Champion1Name, Champion2Name, PatchID),
    FOREIGN KEY (Champion1Name) REFERENCES Champions(Name) ON DELETE CASCADE,
    FOREIGN KEY (Champion2Name) REFERENCES Champions(Name) ON DELETE CASCADE,
    FOREIGN KEY (PatchID) REFERENCES Patch(PatchID) ON DELETE CASCADE
);
```